PROJECT PROPOSAL

Closed-Loop Neuroadaptive VR Platform for **Dopamine Scrolling**



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The Problem

Dopamine scrolling affects every generation, and hits the youngest hardest.

Teenagers
13–17 y/o

21%

severe social media and short-form video addiction

Generation Z

53%

acute dependency or incapacity to stop scrolling

Millennials

46%

chronic attentional dysregulation, compulsive use

Generation X

27–33%

compulsive consumption hurting daily productivity

Baby Boomers

<15%

hooked into dopamine loops by short videos

Negative outcomes: mental distraction, degraded social interaction, sleep disruption, and increased risk of anxiety and depression.

Rationale

- Digital interfaces exploit variable reward schedules and infinite scroll to hijack dopaminergic pathways, the result is a passive, hypnotic state with diminished prefrontal cortex oversight.
- The constant media multitasking and immediate gratification of entertainment-seeking may influence user's cognitive control of attention.

Sources: Neural correlates of short-video application addiction and adolescents' executive functions (2024).

Market Overview

- The global digital mental health market is projected to reach **$45 billion USD** by 2030,
- This demand is further supported by behavioral data showing that **83% of Gen Z** users acknowledge having an unhealthy relationship with their smartphones due to compulsive short-form content consumption.

Sources: Generation Z's Smartphone and Social Media Usage: A Survey. (2019)..

Objectives

General Objective

Develop a closed-loop neuroadaptive VR platform that helps training users to self-regulate impulse dysregulation, attentional fatigue and rapid dopamine-seeking behaviors.

Specific Objectives

- Build a **closed-loop architecture** for real-time streaming of 8-channel EEG data acquired from the AURA device.
- Design an **adaptive, automated Virtual Reality environment** in Unity that dynamically responds to user cognitive states.
- Develop a **real-time signal processing pipeline** for artifact filtering, spectral analysis, and cognitive biomarker extraction.
- Implement a **data management framework** to record, store, and export multimodal telemetry for post-experimental analysis.
- Produce comprehensive **technical documentation** detailing the system architecture, data standards, and pipeline workflows.

Competitor Analysis

Inmersive VR Therapy (PsyTech VR, Psious/Amelia Virtual Care)

Strength: virtual enviroments. **Weakness**: aimed to substance addiction; are guided by a therapist.

App Blockers (Freedom, Opal)

Strength: strong hard-blocking. **Weakness**: punitive design drives churn and ignores underlying dopamine-seeking triggers.

Proposed Solution Advantage

Combines immersive VR presence with closed-loop neuroadaptive triggers, aimed to self-regulate impulse dysregulation and help to restore inhibitory control.

State of the Art

- Scoping reviews show passive media consumption degrades cognitive flexibility and worsens emotional exhaustion.
- Closed-loop physiological monitoring and standalone VR headsets enable high-fidelity detection of attentional fatigue.
- Real-time sensory redirection shifts the field from static app blocking to active neurobehavioral rehabilitation.

Proposed Solution

MIRAI INNOVATION



01

Immersive Detection Engine

Tracks behavioral markers and gaze patterns to identify the transition into the dopamine-scrolling trance.



02

Dynamic Environmental Modulation

Alters the virtual environment's visual and auditory parameters to restore inhibitory control.



03

Active Inhibitory Engagement

Prompts conscious user agency through deliberate interaction tasks that demand intentional effort, strengthening prefrontal inhibitory control.

Closed-loop architecture: Detection → Modulation → Engagement

Development Plan

1 — 2 — 3 — 4

Phase 1

- Software and hardware familiarization
- Conceptual ideation, prototyping, and asset mapping

Phase 2

- Development and building of the core Virtual Reality scene and UI components in Unity
- Initial signal acquisition testing and validation

Phase 3

- End-to-end integration testing between the VR environment, Python processing pipeline, and Data Management System.

Phase 4

- Preparation and execution of live, real-time interactive demonstrations of the neuroadaptive platform.

Technologies Used

MIRAI INNOVATION

Game Engine & VR Framework

Unity (C#) with OpenXR for cross-platform.

Backend & Analytics

Python (Flask, Pandas) for processing telemetry data and session metrics.

Version Control & CI/CD

Git, GitHub, and automated build pipelines.

Hardware Target

VR Headset and AURA device.



INNOVATION FACTOR

A closed-loop therapeutic intervention model within immersive VR.

The first market solution combining real-time EEG signals with immersive VR to self-regulate dopamine scrolling, featuring a telemetry-driven neuroadaptive ecosystem with dynamic environmental shifts tailored for compulsive digital behaviors rather than traditional substance therapies or static apps which forces to stop the use of digital devices.

Target Market

PRIMARY SEGMENT



18–35

years old

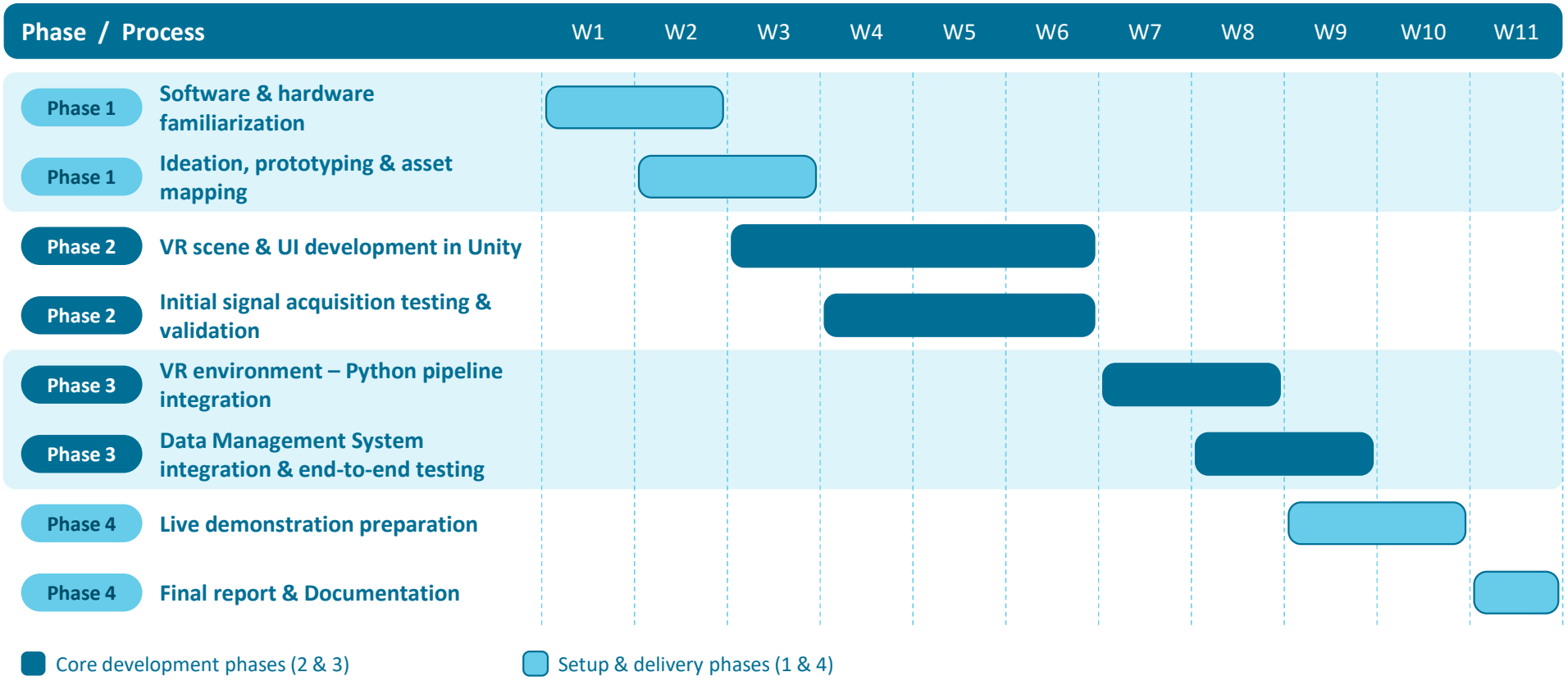
Generation Z and Millennials experiencing high daily screen dependency and attentional fatigue.

SECONDARY SEGMENT



Educational institutions and clinical psychology practices specializing in behavioral addiction and executive function training.

Gantt Diagram



Next Steps

1. Learning and immersing myself in EEG signals, Unity, and VR environment design.
2. Designing the architecture for the data flow between the EEG system and the data management system.
3. Analyze in depth the cerebral reactions and neural signal patterns driven by dopamine scrolling.

References

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